

DEEP RESEARCH / VOXELS3

Keeping GPU Voxel Terrain Ahead of the Player

How production terrain systems preserve coverage, refine LOD by availability, and stay responsive under high-speed traversal.

DECISION	ONE CANONICAL SYSTEM
Availability-driven clipbox refinement	One SDF • one scheduler • one GPU mesher

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Architecture recommendation • no runtime change claimed

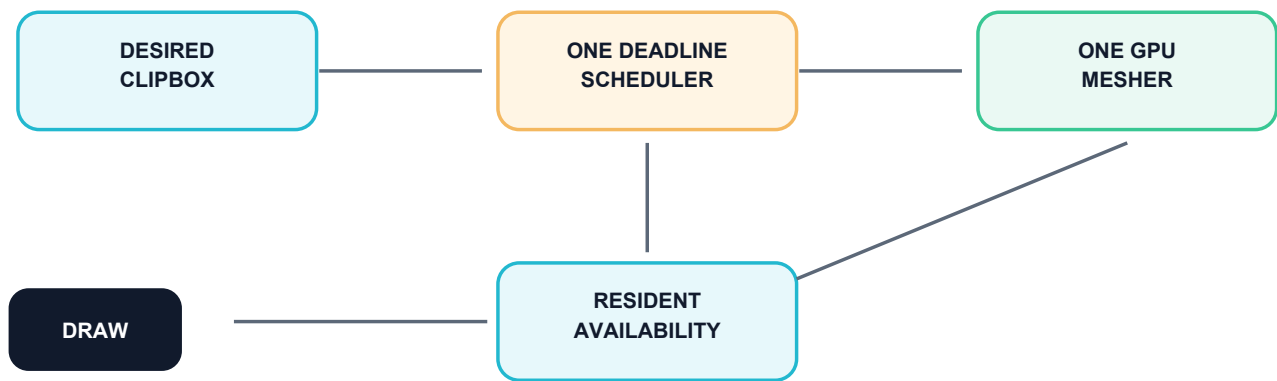
01 / EXECUTIVE ANSWER

Preserve coverage. Let detail lag.

The industry pattern is consistent: the renderer uses the best resident representation, not the most detailed representation merely requested. Under load, terrain gets coarser—not absent.

The recent whole-placement handoff fixed transient holes by withholding a new target until every entering LOD1, LOD2, and transition mesh was ready. That is correct at publication time but too coarse as a scheduling barrier. During the fixed figure-eight run, desired targets are superseded faster than complete placements become publishable, so the player can outrun correct terrain.

Voxels3 should keep one authoritative SDF and one GPU mesher, but separate desired placement from resident drawable placement. Start from coarse resident coverage. Refine a small spatial region only when its complete child set and required 2:1 transition are resident. Retire the parent in the same atomic publication transaction.



Publish only dependency-complete regional refinements

This is not a second fallback renderer. Coarse and fine meshes are derived caches produced by the same mesher from the same SDF. The availability graph only decides which already-canonical cache is safe to draw.

02 / PRODUCTION EVIDENCE

Four patterns recur across games and GPU systems

1

Resident coarse coverage

Geometry Clipmaps crop unavailable fine active regions while valid coarser levels remain. Far Cry 5 begins traversal at the always-resident low LOD and descends only when children are loaded.

2

Incremental movement

Viewer-centered windows shift at snapped boundaries and update only newly exposed strips or L-shaped regions. Motion does not rebuild the full cache.

3

Bounded priority

Nearby, coverage-critical, and dependency-unblocking work beats distant refinement. Budgets cap work per frame; age prevents starvation.

4

Requested $\neq$ resident

Virtual texturing, Nanite, and production terrain distinguish desired fidelity from data that is actually safe to use within a bounded pool.

Transfer limits

Geometry Clipmaps and Far Cry 5 are heightfield systems; their trim geometry and edge morphing cannot replace Transvoxel for caves or arbitrary SDF topology. Nanite is primarily imported/static clustered geometry. Virtual textures are a residency analogy. The transferable contract is availability-driven hierarchy—not their data representation.

Transvoxel remains the 2:1 volumetric topology authority. It does not define streaming priority, temporal publication, or memory ownership.

03 / VOXELS3 EVIDENCE

Correct publication, insufficient convergence

The current candidate achieved the essential correctness result: zero unsafe placement commits and zero missing active resources. Yet the unchanged figure-eight comparison failed frame-tail acceptance and showed material target churn.

318 SUPERSEDED TARGETS	382 ms LOD2 P95 LATENCY	~537 ms OBSERVED STALL	16 / 14 ARENAS VS BASELINE
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Metric	Candidate	Baseline	Decision
Moving CPU p99	2.5284 ms	2.2072 ms	FAIL +0.3212
Moving GPU p99	1.8382 ms	1.5287 ms	FAIL +0.3095
Stationary GPU p99	1.3294 ms	0.9460 ms	FAIL +0.3834
Placement request / commit	571 / 473	—	Lag / churn
Unsafe / missing active	0 / 0	0 required	PASS

Why isolated latency gates mislead

At 2,500 units/s, the measured p95 regular, transition, and LOD2 latencies imply about 253, 538, and 955 units of travel. A target depends on many such resources and also waits behind backlog. The required lead input is p99 target exposure → coverage publication, not one mesh's p95 service time.

Allocator evidence

Arenas 0–12 exhausted all 256 region records while using only roughly 16–19% of vertex capacity and 30–36% of index capacity. Arena growth is primarily descriptor-slot pressure, not geometry bytes. That raises memory and cleanup cost without increasing useful surface output.

04 / CANONICAL ARCHITECTURE

Availability-driven clipbox refinement

Owner / stage	Required behavior	Invariant
Authoritative SDF	Evaluate density and conservative region bounds.	One world truth; edits invalidate the same spatial regions.
Desired placement	Track where the player wants the fixed LOD0/1/2 clipbox to be.	Desired state never becomes drawable by assumption.
One scheduler	Order coverage, transitions, refinement, and prefetch by deadline, dependency, distance, direction, age, and revision.	One bounded work queue; no cascading implementations.
One GPU mesher	Classify/count/emit canonical regular and transition meshes.	Partial internal work is allowed; partial mesh publication is not.
Resident availability	Track which coarse regions, children, and transitions are actually ready.	Rendering selects only dependency-complete state.
Publication	Expose ready children + transition and hide covered parent atomically.	Coverage cannot disappear during refinement.

Publication state per coarse region

- Coarse resident and drawable.
- Refinement requested; children may arrive independently but remain unpublished.
- All children resident and required transition resident.
- One transaction publishes the refinement and retires the covered parent.
- A stale revision can finish GPU work but cannot publish.

The publication unit should be the smallest practical dependency-complete region or slab—not a whole placement. Its exact size is a benchmark decision because tiny commits also carry metadata and draw overhead.

05 / SCHEDULING & MOTION

Stay ahead with deadlines, not radius guesses

Priority	Work	Why
P0	Gameplay collision and immediate safety	Player correctness.
P1	Missing coarse coverage	Prevents an uncovered destination.
P2	Transition dependency that unblocks a ready refinement	Turns completed child work into drawable terrain.
P3	Fine visible refinement	Improves fidelity without risking coverage.
P4	Directional lead and warm prefetch	Builds future headroom.

All five levels are ordering inside one stable scheduler. Age increases effective priority so transitions or distant dependencies cannot starve. Obsolete queued work is canceled before GPU dispatch; submitted work may complete when cancellation is unsafe, but revision guards suppress stale publication.

Motion contract

- Snap the desired anchor with hysteresis; do not rebuild on every camera delta.
- Keep a symmetric coarse-safety bubble for hard turns, reversals, and stationary camera rotation.
- Add directional priority from player velocity and view for extra lead.
- Size lead from measured p99 target-to-coverage latency plus margin.

$$\text{lead distance} \geq \text{supported speed} \times (\text{p99 target-to-coverage} + \text{safety margin})$$

Teleports and editor camera jumps

No finite streamer can guarantee arbitrary instantaneous travel. Use the same system with an explicit destination coarse-ready gate or a visible loading/stand-in state, then refine normally. Do not make an ejected editor camera silently replace the player as the production streaming origin.

First remove work; then accelerate what remains

1. Conservative SDF surface rejection

LOD1 and LOD2 currently enumerate every vertical region in a 16×16×16 cache even though the procedural surface occupies a bounded height range. Add an SDF-owned conservative density or height interval. Regions proven entirely air or solid should never enter GPU extraction or consume geometry records.

The bound must reuse the authoritative generator. A duplicated approximation is forbidden: a false rejection creates the same holes this work is meant to prevent.

2. Rebalance arena records

Measure descriptor slots and vertex/index bytes independently. The smallest candidate is a tested increase in records per slab while retaining byte limits and indirect-draw semantics. If that harms scans or uploads, separate compact descriptors from geometry ranges inside the same mesher.

3. Investigate GPU compaction only after measurement

The installed s&box; 26.09.01a API exposes structured/append buffers, append-counter reset and structure-count copy, indirect compute dispatch, and indexed indirect drawing. A scan/append/indirect path is technically plausible. API presence does not prove a safe allocator, and prior compaction/device-loss evidence makes this a later isolated research slice.

Current	Near-term	Only if still measured
GPU count → scalar CPU readback → CPU arena allocation → GPU emit	Reject empty/solid work; prioritize by deadline; publish regionally	GPU scan/append compaction and indirect emit/draw

07 / DESIGN DISPOSITION

What transfers—and what does not

Idea	Decision	Reason
Resident coarse coverage + availability refinement	ADOPT	Core no-hole publication contract.
Incremental clipmap movement	ADAPT	Use fixed volumetric LOD0/1/2 and Transvoxel.
Priority / time-budgeted streaming	ADOPT	One deadline scheduler with aging.
Conservative SDF range rejection	ADOPT	Canonical broad-phase before extraction.
Whole-placement atomic barrier	REJECT	Slowest dependency blocks the whole target.
Second heightfield or CPU fallback	REJECT	Competing terrain truth and duplicated features.
Nanite as direct voxel solution	REJECT	Static clustered geometry is not live SDF extraction.
GigaVoxels raycast renderer	REJECT	Useful streaming evidence, incompatible renderer path.
Raise dispatch budget blindly	REJECT	Can turn lag into frame-tail stutter.
Velocity-only forward cone	REJECT	Unsafe on reversals and hard turns.
Generalized N-level hierarchy	DEFER	Fixed three levels meet the current product slice.

One robust system does not mean one giant commit. It means one authority, one scheduler, one mesher, and one dependency-safe publication protocol operating at the granularity that can actually converge.

08 / DELIVERY PATH

Measure coverage—not just mesh service time

Keep the canonical figure-eight v1 parameters unchanged. Add measurements to the production path; do not create a smaller smoke scenario and compare it as equivalent.

New required telemetry

Coverage	Scheduler	Resources
p99 target → first coarse coverage	Deadline misses / worst lateness	Arena record-slot utilization
p99 target → requested refinement	Obsolete requests canceled	Vertex / index byte utilization
Maximum target lag (chunks + seconds)	Dependency-blocked publications	Candidate / indirect overflows
Coarse-fallback frames + world area	Air / solid classification counts	Peak, resident, growth, cleanup

Recommended implementation order

- 01 Instrument target-to-coverage latency, refinement lag, deadline misses, cancellation, and classification.
- 02 Add canonical conservative SDF region classification; skip guaranteed air/solid extraction.
- 03 Replace whole-placement readiness with dependency-safe regional coarse-to-fine publication.
- 04 Unify category FIFO behavior into one deadline-aware scheduler.
- 05 Correct arena record packing using separate slot and byte evidence.
- 06 Only then prototype GPU append/scan/indirect compaction if readback remains material.

Acceptance requires both: zero uncovered safety-region terrain and a comparable figure-eight performance pass.

09 / PRIMARY SOURCES

Evidence and transfer boundaries

- [Geometry Clipmaps](#) — [Losasso & Hoppe](#) — Nested viewer-centered grids, incremental updates, fine-region cropping under pressure.
- [GPU-Based Geometry Clipmaps](#) — [NVIDIA GPU Gems 2](#) — GPU-oriented clipmap implementation; heightfield-specific.
- [Terrain Rendering in Far Cry 5](#) — [Ubisoft / GDC](#) — Requested vs loaded state; resident coarse root; GPU traversal, LOD, culling, stitching.
- [Streaming in Sunset Overdrive's Open World](#) — [Insomniac / GDC](#) — Speed-derived budgets, initialization cost, allocation, teleport constraints.
- [The Transvoxel Algorithm](#) — [Eric Lengyel](#) — Authoritative local 2:1 volumetric transition topology.
- [Godot Voxel performance + smooth terrain](#) — Priority/time budgets and conservative SDF clipping; CPU-oriented transfer limits.
- [GPU procedural terrain](#) — [NVIDIA GPU Gems 3](#) — GPU block generation, bounded pools, remembered empty blocks.
- [CUDA Samples: Marching Cubes](#) — [NVIDIA](#) — Prefix-scan compaction evidence.
- [Sampler Feedback](#) — [Microsoft DirectX specification](#) — Desired vs actually resident mip model.
- [Nanite + virtual texture pools](#) — [Epic Games](#) — GPU hierarchy, bounded residency, overflow/thrash diagnostics; not live voxel extraction.
- [Worlds Part I](#) — [Hello Games](#) — Official dual-Marching-Cubes performance claims; no scheduler details.
- [GigaVoxels](#) — [Crassin et al.](#) — Feedback-guided bounded sparse-volume streaming; different renderer.

Project evidence

Fixed figure-eight candidate and baseline measurements are recorded in `Docs/ValidationResults.md`. The full adopt/adapt/reject record is in `Docs/Research/GpuVoxelTerrainStreaming.md`. Installed s&b;box; API availability was verified against `Sandbox.Engine.dll` and `Sandbox.Engine.xml` from editor 26.09.01a on 2026-09-03.